

Stat 3113:

Lesson 11



Discrete Random
Variables

Definition:

- A **probability mass function (pmf)** of a discrete random variable is defined as:

$$p(x) = P(X = x) \text{ for all } x. \in X(S) \text{ or } X(\Omega) \text{ its domain/support.}$$

- The pmf is often written in a chart format like this:

Top row →

x	x1	x2	.	.	.	ow
p(x)	p(x1)	p(x2)	.	.	.	0

← Bottom row.

- Top row: x followed by the possible values of the random variable.
- Bottom row: $p(x)$ followed by $P(X = x_i)$ for each value in row above.

Remark: “ow” means otherwise; it represents all the real numbers for which the random variable X has a 0 probability.

Example 1: Suppose 6 boxes of parts are being shipped, and it is known that the number of defects in each box is as follows:

Box	1	2	3	4	5	6
# def	0	2	0	1	2	0

Suppose a box is randomly selected. Let $X = \#$ defects in the chosen box. Find the pmf of X .

- $X(s) = \{0, 1, 2\}$
- pmf: Equally likely outcomes (The boxes are not distinguishable)

x	0	1	2	ow
$p(x)$	$P(X=0) = \frac{\text{\# boxes with 0 defects}}{\text{Total \# of boxes}} = \frac{3}{6} = \frac{1}{2}$ ≥ 0	$\frac{1}{6}$ ≥ 0	$\frac{2}{6} = \frac{1}{3}$ ≥ 0	0 ≥ 0

Total

$$\frac{3}{6} + \frac{1}{6} + \frac{2}{6} = \frac{6}{6} = 1$$

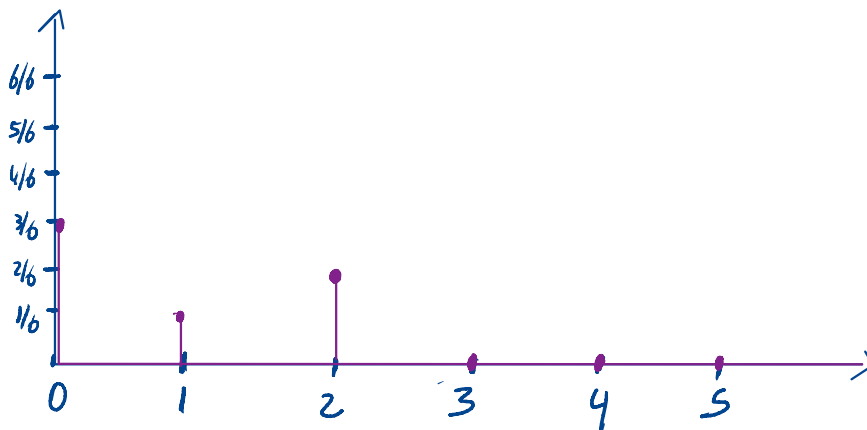
Properties of the pmf:

1. $p(x) \geq 0$

2. $\sum_{\text{all } x_i} p(x_i) = 1$

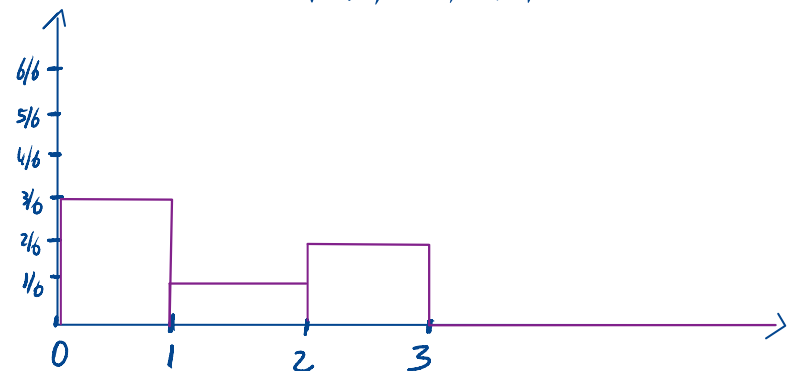
- Graph of pmf above as a function and then as a histogram:

As a function



As a histogram

If we split the values into the intervals:
- $[0, 1)$, $[1, 2)$, $[2, 3)$



Example 2: Find the following probabilities using above example:

a. $P(0 \leq X \leq 1)$

$$\begin{aligned} P(0 \leq X \leq 1) &= P(X=0) + P(X=1) \\ &= \frac{3}{6} + \frac{1}{6} = \frac{4}{6} = \frac{2}{3}. \end{aligned}$$

b. $P(0 < X \leq 1)$

$$\begin{aligned} P(0 < X \leq 1) &= P(X=1) \\ &= \frac{1}{6} = \frac{1}{6}. \end{aligned}$$

c. $P(X > 0)$

$$\begin{aligned} P(X > 0) &= 1 - P(X \leq 0) \\ &= 1 - P(X = 0) \\ &= 1 - \frac{1}{2} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} P(X > 0) &= P(X = 1) + P(X = 2) \\ &= \frac{1}{6} + \frac{2}{6} \\ &= \frac{3}{6} = \frac{1}{2} \end{aligned}$$

d. $P(X \geq 1.5)$

$$\begin{aligned} P(X \geq 1.5) &= P(X = 2) \\ &= \frac{1}{3} \end{aligned}$$

e. $P(X = 2)$

$$IP(X=2) = \frac{1}{3}$$

($IP(X=2) = 0$ if X was a continuous random variable)

f. $P(X = 1.5)$

$$IP(X=1.5) = 0 \text{ since } 1.5 \notin X(S).$$

g. $P(1.5 \leq X < 2.5)$

$$IP(1.5 \leq X < 2.5) = IP(X=2) = \frac{1}{3}$$

Definition: The **cumulative distribution function (cdf)** of a discrete rv X with pmf $p(x)$ is defined as:

$$F(x) = P(X \leq x) = \sum_{y \leq x} p(y) \quad \text{for all } x.$$

Note:

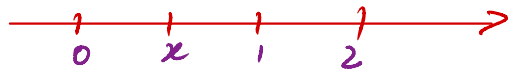
- Since the cdf is $P(X \leq x)$, the method to obtain it involves adding all the pmf values for the X values $\leq x$.
- This means the cdf will “jump” at values where the pmf had non-zero values because these non-zero probabilities are added.

Example 3:

- Obtain the cdf for the above example.

x	0	1	2	ow
p(x)	$\frac{1}{2}$	$\frac{1}{6}$	$\frac{1}{3}$	0

- $F: \mathbb{R} \longrightarrow [0, 1]$
 $x \longmapsto F(x) = P(X \leq x)$

Let $x \in \mathbb{R}$, and we consider the following cases: 

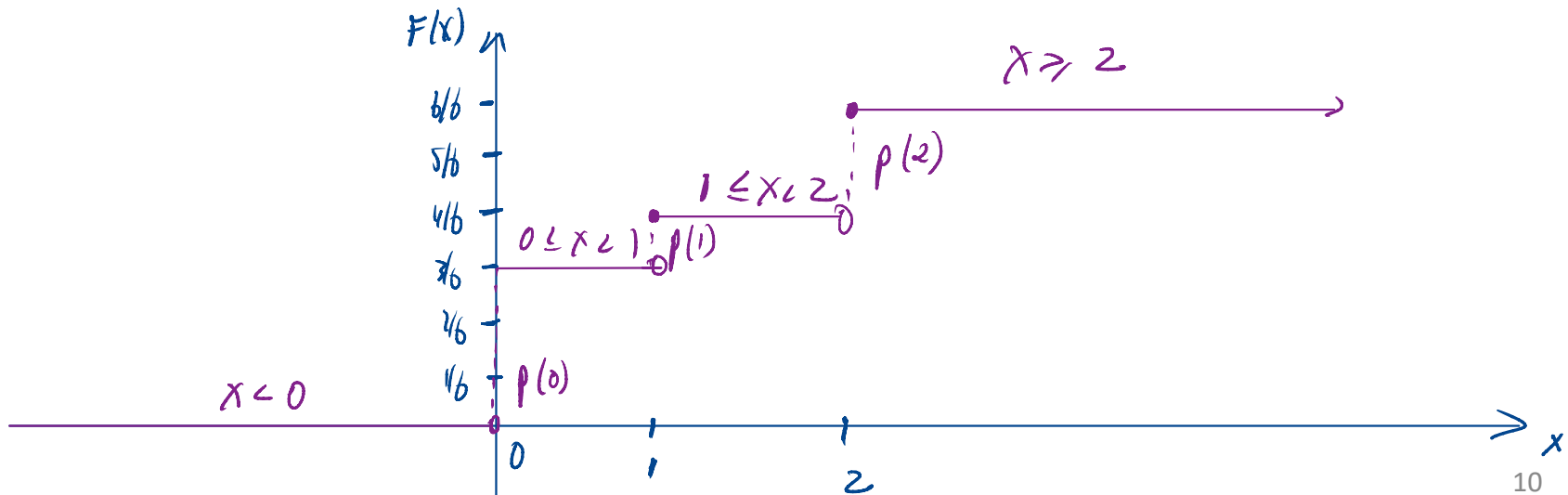
- If $x < 0$, $F(x) = P(X \leq x) = 0$ because X doesn't take on any value before 0.
- If $0 \leq x < 1$, $F(x) = P(X \leq x) = P(X=0) = \frac{3}{6}$
- If $1 \leq x < 2$, $F(x) = P(X \leq x) = P(X=0) + P(X=1) = \frac{4}{6}$
- If $x \geq 2$, $F(x) = P(X \leq x) = P(X=0) + P(X=1) + P(X=2) = 1$.

Continued:

2

$$F(x) = \begin{cases} 0 & , \text{ if } x < 0 \\ 3/6 & , \text{ if } 0 \leq x < 1 \\ 4/6 & , \text{ if } 1 \leq x < 2 \\ 1 & , \text{ if } x \geq 2 \end{cases}$$

- Graph of the cdf above.



Example 4: Consider the following cdf of a discrete random variable X :

$$F(x) = \begin{cases} 0, & x < 1 \\ 0.25, & 1 \leq x < 3 \\ 0.60, & 3 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$$

Find the pmf of the rv X .

- $X(s) = \{1, 3, 4\}$
- We will make use of the formula: $p(x_i) = \begin{cases} F(-\infty) = 0 \\ F(x_i) - F(x_{i-1}) \\ F(\infty) = 1 \end{cases}$

x_i	1	3	4	ow
$p(x_i)$	$p(1) = F(1) - F(-\infty)$ $= 0.25 - 0$ $= 0.25$	$p(3) = F(3) - F(1)$ $= 0.60 - 0.25$ $= 0.35$	$p(4) = F(4) - F(3)$ $= 1 - 0.60$ $= 0.4$	0

$$\sum_{i=1}^3 p(x_i) = 1.$$

Definition:

1. For a discrete rv X , the **expected value of X** is:

$$E(X) = \mu_X = \mu = \sum_{\text{all } x} x p(x)$$

↑
expected value

↘ Introduction

Example 5: Find the $E(X)$ for the previous example (from 3.2 notes and see below):

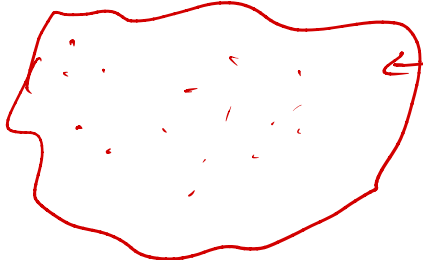
Box	1	2	3	4	5	6
# def	0	2	0	1	2	0

x	0	1	2	ow
p(x)	$1/2$	$1/6$	$1/3$	0

$$\begin{aligned} E[X] &= (0 \times \frac{1}{2}) + (1 \times \frac{1}{6}) + (2 \times \frac{1}{3}) \\ &= 0 + \frac{1}{6} + \frac{2}{3} = \frac{1}{6} + \frac{4}{6} = \frac{5}{6} \approx 1 \end{aligned}$$

Interpretation: The expected number of defects in a randomly selected box is roughly 1.

Note: The **expected value** of a rv is the same thing as the **mean of the population**, $\mu = \mu_X$.

•  $X := \text{variable}$
↓
Data = {0, 12, 7, 4, ...}
↳ No observation.
↓
 $\mu_X = \frac{1}{n} \sum_{i=1}^n x_i$, S^2 , σ .
What if we have instead knowledge about the distribution of X , that's its **pdf**?
↳ Expected Value of X .

2. For a discrete rv X , the **variance of X** is: $V(X) = E[(X - E(X))^2]$

Calculation Formula for Variance:

$$V(X) = E(X^2) - (E(X))^2 \quad \text{where} \quad E(X^2) = \sum_{\text{all } x} x^2 p(x)$$

↓ Proof below! ⊛

Note: There are other notations for the variance of X . They are:

$$V(X) = \sigma_X^2 = \sigma^2$$

3. The **standard deviation of X** is: $\sigma_X = \sigma = \sqrt{V(X)}$

Example 6: Find $V(X)$ and σ_X for the above example.

$$V(X) = E[X^2] - (E[X])^2$$

$$E[X^2] = \sum_{\text{all } x} x^2 p(x)$$

$$= \left(0^2 \times \frac{1}{2}\right) + \left(1^2 \times \frac{1}{6}\right) + \left(2^2 \times \frac{1}{3}\right)$$

$$= \frac{1}{6} + \frac{4}{3} = \frac{9}{6} = \frac{3}{2} = 1.5$$

$$\begin{aligned} \bullet V_{\text{cal}}(X) &= V(X) = \frac{3}{2} - \left(\frac{5}{6}\right)^2 \\ &= \frac{3}{2} - \frac{25}{36} \\ &= \frac{54 - 25}{36} \end{aligned}$$

$$V(X) = \frac{29}{36} \approx 0.81$$

$$\bullet \sigma_X = \sqrt{V(X)} = \sqrt{0.81} \approx 0.9$$

Useful Results: $\bar{E}X: ^\circ C = 5/9(^{\circ}F - 32) \rightarrow \text{Motivation?}$

$$1. E[g(x)] = \sum_{\text{all } x} g(x) p(x)$$

$$2. E(aX + b) = aE(X) + b$$

④ Proof

$$E[(X - E[X])^2] = E[X^2 - 2XE[X] + E[X]^2] = E[X^2] - 2E[X]E[X] + E[X]^2 = E[X^2] - E[X]^2.$$

$$3. V(aX + b) = a^2 V(X)$$

Example 7: Find $E(3X + 2)$ and $V(3X + 2)$ for the above example.

$$\begin{aligned} \bullet E[3X + 2] &= 3E[X] + 2 \\ &= 3 \cdot \frac{5}{6} + 2 \\ &= \frac{5}{2} + \frac{4}{2} = \frac{9}{2} \end{aligned}$$

$$\begin{aligned} \bullet V(3X + 2) &= 3^2 V(X) \\ &= 9 \cdot \frac{29}{36} \\ V(3X + 2) &= \frac{29}{4} \end{aligned}$$

Example 8:

The manager of a large computer network has developed a probability distribution of the number of interruptions to the network each day (X). The pmf she developed is as follows:

x	0	1	2	3	4	5	6	ow
p(x)	0.32	0.35	0.18	0.08	0.04	0.02	0.01	0

$$\sum_{\text{all } x_i} p(x_i) = 1.$$

Compute the following:

a. The probability the number of network interruptions is more than 3 for a given day.

$$\begin{aligned} IP(X > 3) &= IP(X=4) + IP(X=5) + IP(X=6) = p(4) + p(5) + p(6) \\ &= 0.04 + 0.02 + 0.01 \\ &= 0.07. \end{aligned}$$

b. The probability the number of network interruptions is less than 2 for a given day.

$$\begin{aligned} IP(X < 2) &= IP(X=0) + IP(X=1) = p(0) + p(1) \\ &= 0.32 + 0.35 \\ &= 0.67 \end{aligned}$$

c. The probability the number of network interruptions is at least 5 for a given day.

$$\begin{aligned} IP(X \geq 5) &= IP(X=5) + IP(X=6) = p(5) + p(6) \\ &= 0.02 + 0.01 \\ &= 0.03 \end{aligned}$$

d. The probability the number of network interruptions is between 2 and 5 (inclusive) for a given day.

$$\begin{aligned} IP(2 \leq X \leq 5) &= 1 - [IP(X < 2) + IP(X > 5)] = p(2) + p(3) + p(4) + p(5) \\ &= 1 - [0.67 + 0.01] = 0.32 \\ &= 0.18 + 0.08 + 0.04 + 0.02 \\ &= 0.32 \end{aligned}$$

e. The probability the number of network interruptions is between 2 and 5 (non-inclusive) for a given day.

$$\begin{aligned} IP(2 < X < 5) &= p(3) + p(4) \\ &= 0.08 + 0.04 \\ &= 0.12 \end{aligned}$$

f. Compute the mean (expected value) number of network interruptions per day.

$$\begin{aligned} \mu = E[X] &= \sum_{\text{all } x} x p(x) = (0 \times 0.32) + (1 \times 0.35) + (2 \times 0.18) + (3 \times 0.08) + (4 \times 0.04) \\ &\quad + (5 \times 0.02) + (6 \times 0.01) \\ &= 1.27 \end{aligned}$$

g. Compute the standard deviation of the number of network interruptions per day.

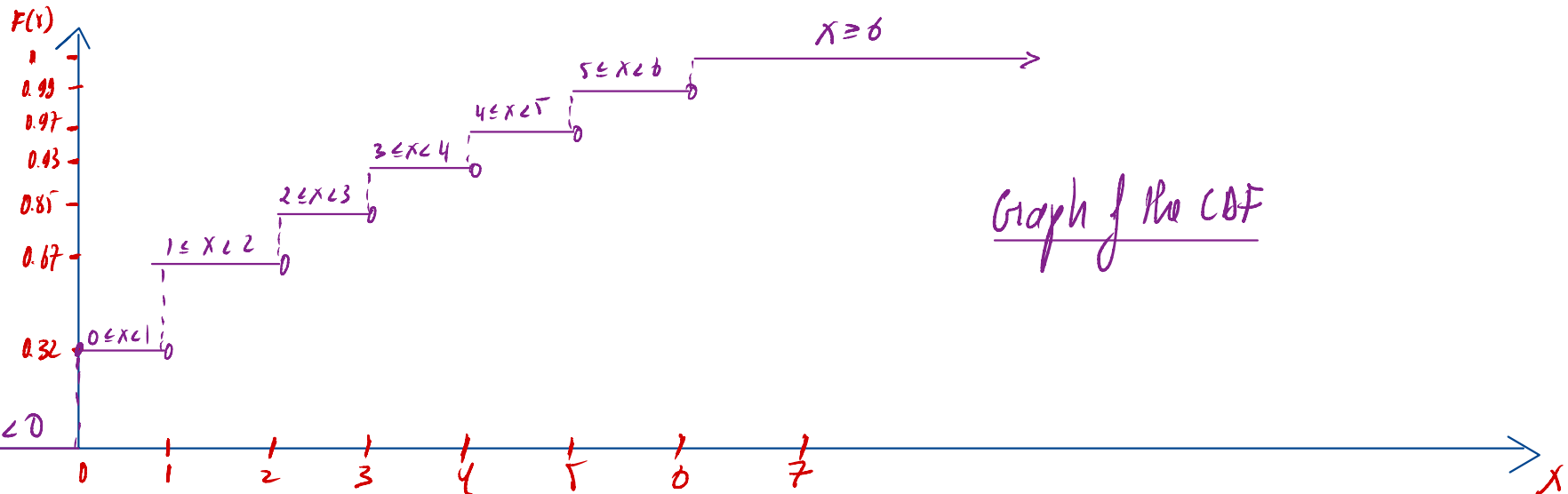
$$\begin{aligned} \bullet \ E[X^2] &= \sum_{\text{all } x} x^2 p(x) = (0^2 \times 0.32) + (1^2 \times 0.35) + (2^2 \times 0.18) \\ &\quad + (3^2 \times 0.08) + (4^2 \times 0.04) \\ &\quad + (5^2 \times 0.02) + (6^2 \times 0.01) \\ &= 3.29 \end{aligned}$$

$$\begin{aligned} \bullet \ V(X) &= E[X^2] - E[X]^2 = 3.29 - 1.27^2 \\ &= 1.68 \end{aligned}$$

$$\bullet \ \sigma_X = \sqrt{V(X)} = 1.30.$$

h. Obtain the cdf.

$$F(x) = \begin{cases} 0 & , x < 0 \\ 0.32 & , 0 \leq x < 1 \\ 0.67 & , 1 \leq x < 2 \\ 0.85 & , 2 \leq x < 3 \\ 0.93 & , 3 \leq x < 4 \\ 0.97 & , 4 \leq x < 5 \\ 0.99 & , 5 \leq x < 6 \\ 1 & , x \geq 6 \end{cases}$$



Reading

Chapter 3 – Section 3.3



Next Topic



The **Binomial** Probability Distribution



Chapter 4 - **Sections 4.1, 4.2**